

Experiment	<i>Lab 5: Supersonic Flow Visualization (Supersonic Wind Tunnel)</i>
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Summary

Supersonic Flow Visualization (Supersonic Wind Tunnel) lab (SWT) analyzes flow through a converging to diverging nozzle to find flow characteristics like pressure distribution, nozzle pressure ratio (NPR), shock strength, mass flow rate, and thrust. By using static pressure measurements along the nozzle wall and converting to absolute pressure, with a 95% confidence interval, the pressure distribution displayed decrease in pressure followed by sharp increase at the flow separation point (ports 11 through 12). The NPR = 2.0686, showing overexpanded flow. The pressure ratio for shock strength = 1.5391, which is consistent with compressible theory. After assuming choked flow at the throat, the $\dot{m} = 0.2392 \frac{kg}{s}$ and thrust or $\tau = 86.4328 N$. Overall, the results confirm nozzles used in an overexpanded regime with internal shock and flow separation are consistent for Schlieren visualization and pressure distribution .

Aim & Objectives

The aim of this experiment was to determine overexpanded compressible flow in a converging-to-diverging nozzle using a supersonic wind tunnel, Schlieren visualization, and wall pressure ports. Objectives include operating and understanding a supersonic blow-down wind tunnel, handling Schlieren imaging for the nozzle calculating NPR, mass flow, and thrust with the measured static pressure from ports.

Equipment Identification

The major components of the supersonic wind tunnel are shown below in Figure 1.

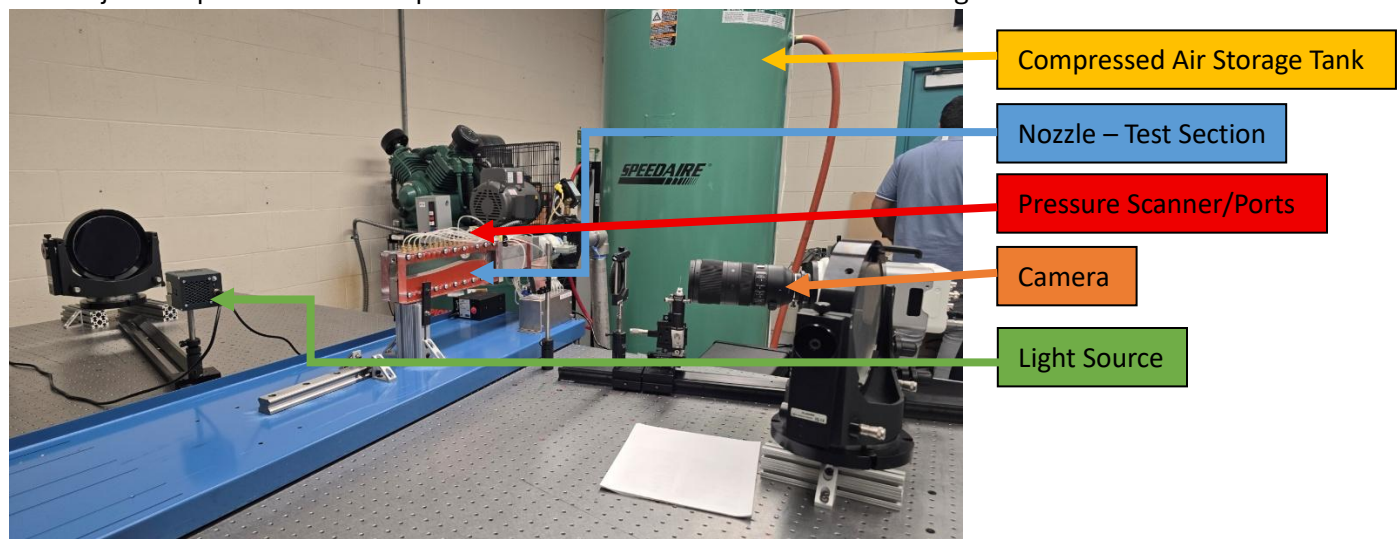


Figure 1: Supersonic wind tunnel as found in the classroom.

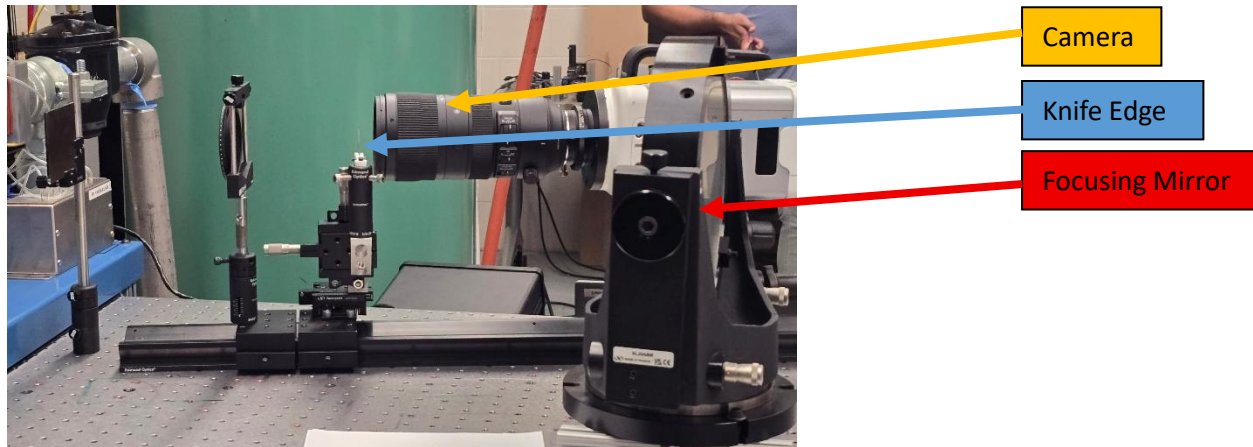


Figure 2: Close-up of the Schlieren system.

Uncertainty Analysis

By selecting the steady-state plateau from the time, the mean static pressure and standard deviation was found. The method used to calculate uncertainty is given that all static pressure measurements have an uncertainty of 95%. The mean pressure data for the analysis was found by reducing the data over the steady state portions. Uncertainty used a confidence interval of 95% of the mean from the standard deviation of the samples divided by the square root of the sample size multiplied by 1.96, as shown by error bars on the wall static pressure distribution figure. The lab handout stated that $p_{lab} = 101760 \text{ Pa}$ and $T_0 = 296.15 \text{ K}$ for the mass flow calculations. The uncertainty for these values is arbitrary as we treat these values as fixed and given. All computations were done in MATLAB (Appendix A).

Results & Discussion

The Schlieren image (Figure 3 below) shows the flow regime with a normal shock in the diverging section. The flow regime is identifiable by the change in color gradient, with light/dark lines in the middle of the image corresponding to shock waves. Further along the stream of the nozzle, the flow transitions from supersonic to subsonic, whereas upstream the flow seems relatively uniform. This indicates that the flow is in the overexpanded flow regime where the exit pressure is lower than the ambient pressure.

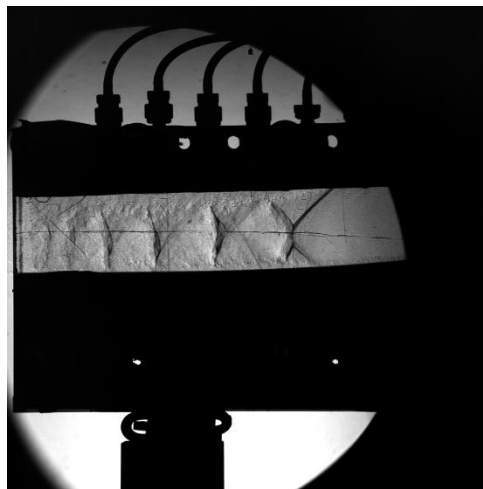


Figure 3: Schlieren image of the test section.

As given by the lab instructions, ambient pressure and stagnation temperature are respectively $p_{lab} = 101760 \text{ Pa}$ and $T_0 = 296.15 \text{ K}$. The values are used in calculations in the following paragraphs to determine NPR, mass flow rate, and thrust.

Wall Static Pressure Distribution

Figure 4 below displays the pressure distribution from the pressure ports on the nozzle and converted from gauge to absolute pressure via the ambient pressure given. The figure is plotted as a function along the nozzle with the ports as reference for distance. The pressure decreases through the converging section and reaches a minimum at the throat, then in the diverging section the pressure keeps decreasing until the flow is supersonic. A sharp rise in pressure is found between ports 11 and 12, indicating separation shock. This is consistent with the transition from supersonic to subsonic flow in this nozzle.

The distance of the ports can be found commented out in the MATLAB code in Appendix A. Error bars in this graph are at a 95% confidence interval (CI). The scaling of the y-axis may give the perception that the CI is incorrect, however, the confidence interval is calculated in Appendix A below.

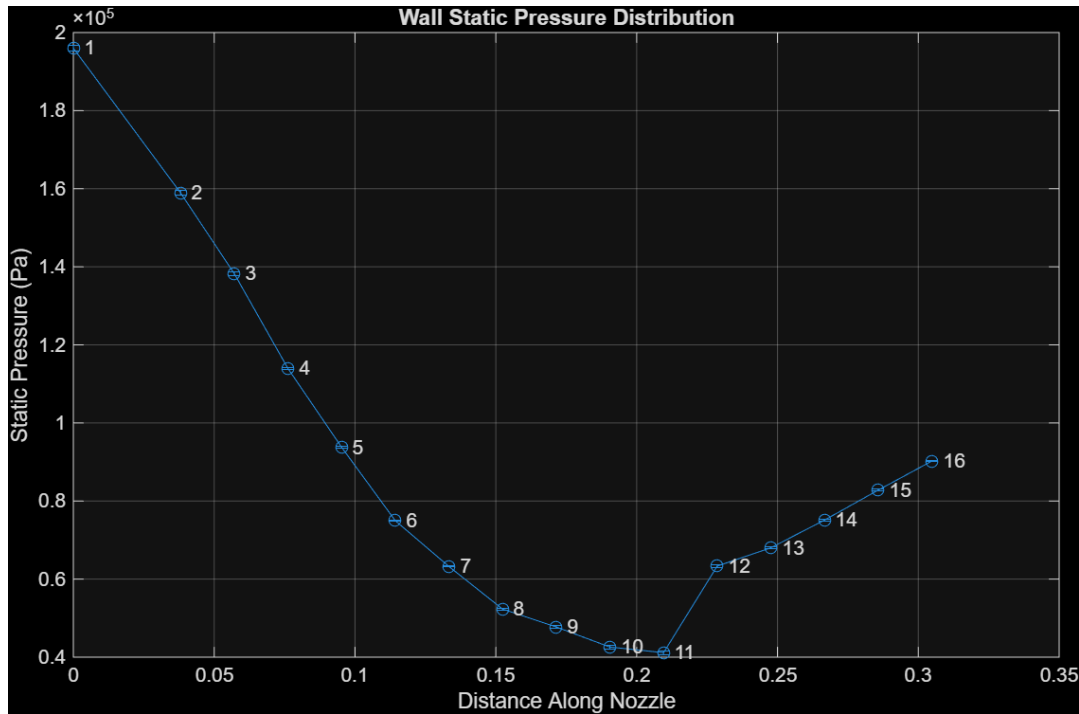


Figure 4: Static pressure versus distance along the nozzle with 95% error bars and port numbers.

Nozzle Pressure Ratio (NPR)

The $NPR = 2.0686$ was calculated using $NPR = \frac{p_0}{p_{lab}}$, where the stagnation pressure (p_0) was found using the static pressure in the subsonic region with isentropic flow relations and given area ratio at $\frac{A}{A_{th}} = 1.9167$. The value of NPR indicated that the nozzle is in overexpanded conditions where exit pressure is lower than ambient pressure. This is shown consistently with the pressure distribution and Schlieren image.

Shock Strength

The strength of the separation shock was calculated using the ratio of pressure immediately down/upstream of the shock. Shock Strength equaled 1.5391. This is consistent with normal shock values.

Mass Flow Rate

Assuming choked flow at the nozzle's throat, the mass flow rate was calculated using

$\dot{m} = A_t \frac{p_0}{\sqrt{T_0}} \sqrt{\frac{\gamma}{R}} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$ and the given stagnation conditions. Mass flow rate is calculated to be 0.2392 kg/s. This value shows that the shock does not affect the upstream mass flow rate as the downstream mass flow rate is independent due to the choked nozzle.

Ideal Thrust

The thrust was calculated with the equation... $\tau = \dot{m}(V_e - V_\infty) + A_e(p_e - p_\infty)$. After neglecting $A_e(p_e - p_\infty)$ as instructed by the TA, the value of thrust equaled 86.4328 N.

The conclusions of this lab support the idea that the nozzle is an overexpanded flow regime. This is supported by the pictures, flow visualization, data, and graphs demonstrating the characteristics of compressible flow in a converging-diverging nozzle.

References

AE315_SWT_(3.21.2026).pdf

ExpAero_Supersonic.pdf

Appendix

```
% AE315 DB15 Group 15.2: Lab 5
% Clancy Crawford Student ID:2654296
% 4/1/2026

clear;
close all;
clc;

% Givens
p_inf = 101760; % Pa
t_inf = 296.15; % K
gamma = 1.4;
R = 287; % J/(kg*K)

data = readmatrix('15DB_Cpdata.csv'); % psig
p_gauge = data(:, 2:17); % psig
p_gauge = p_gauge * 6894.76; % Pa

% finding plateau
%figure
%plot(data(:, 1), data(:, 5))
p_plateau = p_gauge(9570:12409, :);
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```

p_ss_abs = p_plateau + p_inf; % Pa
p_mean = mean(p_ss_abs, 1);
p_std = std(p_ss_abs, 0, 1);
p_err = 1.96 * p_std ./ sqrt(size(p_ss_abs, 1));

p_x_loc = [0 1.5 2.25 3.00 3.75 4.50 5.25 6.00 6.75 7.50 8.25 9.00 9.75 10.50 11.25 12.00]; %
pressure port locations
p_x_loc = p_x_loc * 0.0254; % m

% plot
figure;
errorbar(p_x_loc, p_mean, p_err, 'o-')
grid on
xlabel('Distance Along Nozzle (m)')
ylabel('Static Pressure (Pa)')
title('Wall Static Pressure Distribution')
% Add port number labels
for i = 1:length(p_x_loc)
    text(p_x_loc(i), p_mean(i), sprintf(' %d', i))
end

% using comp flow calc with A/Ath = 1.9167 and gamma = 1.4
% so M1 = 0.32097, then p/p0 = 0.9311
M1 = 0.32097827;
p_p0 = 0.93110244;
p0_p1 = 1/p_p0;
p_inlet = p_mean(1); % inlet port
p0 = p0_p1 * p_inlet; % Pa

% NPR
NPR = p0/ p_inf;
fprintf('NPR: %0.4f\n', NPR)

% seperation occurs between port 11 and 12
p_exit = p_mean(11);
p_a_shock = p_mean(12);
ss = p_a_shock / p_exit;
fprintf('Shock Strength: %0.4f\n', ss);

% mass flow rate
A_star = 0.75 * (0.0254^2); % m^2
m_rate = A_star * p0 / sqrt(t_inf) * sqrt(gamma/R) * (2/(gamma+1))^(gamma+1/(2*(gamma-1))); %
kg/s
fprintf('Mass flow rate: %0.4f kg/s\n', m_rate);

% @ exit
M_exit = sqrt((2/(gamma-1)) * ((p0/p_exit)^(gamma-1/gamma) - 1));
T_exit = t_inf / (1 + (gamma-1)/2 * M_exit^2);
a_exit = sqrt(gamma * R * T_exit);
V_exit = M_exit * a_exit; % m/s

% @ inlet

```

```

T1 = t_inf / (1 + (gamma-1)/2 * M1^2);
a1 = sqrt(gamma * R * T1);
V1 = M1 * a1; % m/s

% thrust
thrust = m_rate * (V_exit - V1); % N
fprintf('Ideal Thrust: %0.4f N\n', thrust);

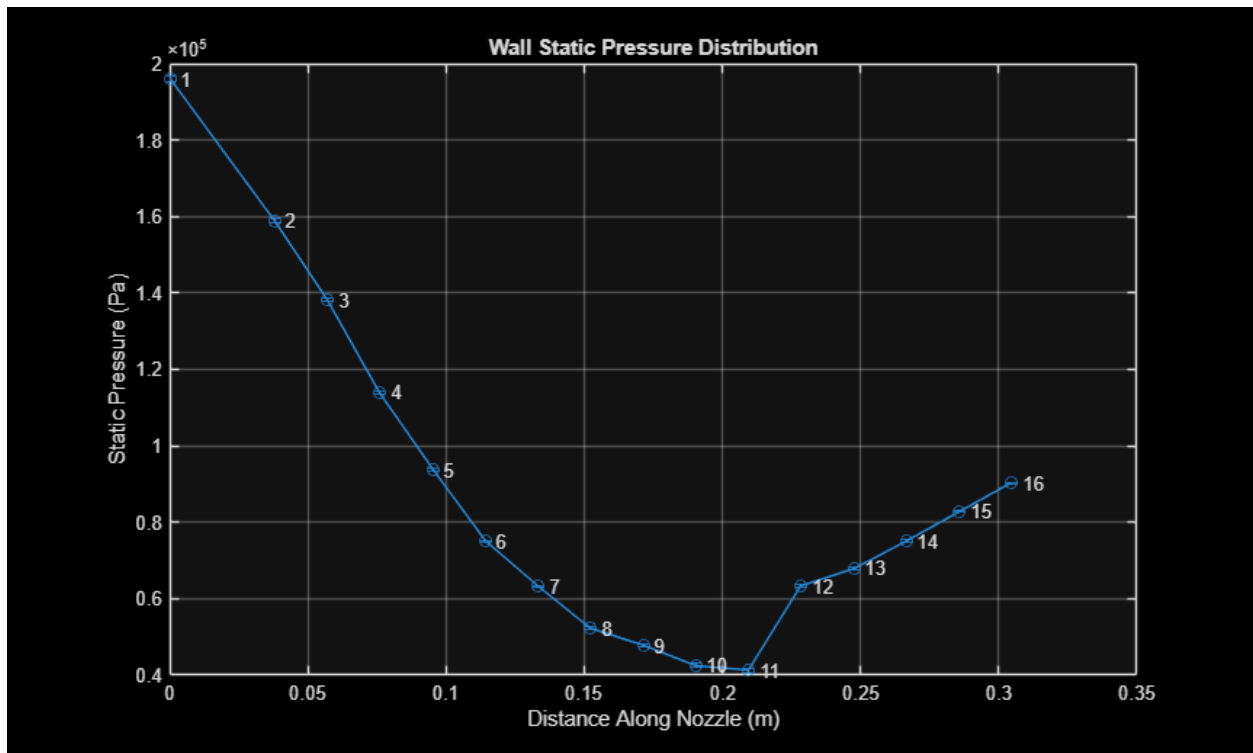
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NPR: 2.0686

Shock Strength: 1.5391

Mass flow rate: 0.2392 kg/s

Ideal Thrust: 86.4328 N



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